Effect of chronic respiratory acidosis on calcium metabolism in the rat

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Chronic metabolic acidosis typically results in hypercalciuria and negative calcium balance. The impact of chronic respiratory acidosis on calcium metabolism has been less well studied. To address this issue, metabolic balance and static bone histomorphometric data were obtained during a 14-day exposure of rats to 10% CO_2 (blood pH 7.33, Pa_{CO_2} 83 mm Hg) and were compared with pair-fed controls. All rats were fed a 0.8% calcium diet. Urinary calcium excretion (mg/period, mean ± SEM) was increased during both week 1 and week 2 (16 ± 3 vs 9 ± 1 and 16 ± 2 vs 9 ± 1, CO_2 group vs controls, respectively [$p < 0.05$]). Net intestinal calcium absorption (intake minus fecal excretion) was increased throughout the period of hypercapnia (week 1, 213 ± 19 mg vs 135 ± 15 mg; week 2, 135 ± 16 mg vs 43 ± 14 mg; and cumulatively, 344 ± 27 mg vs 178 ± 20 mg, CO_2 group vs controls [$p < 0.01$]). As a consequence of the marked increment in intestinal calcium absorption during hypercapnia, mean net calcium balance was more positive than that of controls throughout the study (week 1, 197 ± 18 mg vs 126 ± 15 mg; week 2, 120 ± 15 mg vs 34 ± 15 mg; and cumulatively, 317 ± 25 mg vs 159 ± 20 mg, CO_2 group vs controls, respectively [$p < 0.01$]). There were no significant differences in calcium intake, plasma total calcium, immunoreactive parathyroid hormone, 25-hydroxyvitamin D, or creatinine clearance between the two groups. 1,25 dihydroxyvitamin D, however, was decreased in the CO_2 group versus controls, 45 ± 6 pg/ml versus 82 ± 9 pg/ml, respectively ($p < 0.05$). In the CO_2 group, no significant changes in endosteal bone cell activity were present to suggest increased bone resorption or decreased bone formation. These results indicate that gut net absorption of calcium is enhanced during chronic respiratory acidosis in the rat and results in significant calcium retention. Hypercalciuria also occurs, which might reflect, at least in part, hyperabsorption of dietary calcium or diminished renal tubular calcium absorption. The mechanism of these changes in calcium balance during chronic respiratory acidosis and their impact on dynamic bone metabolism require further study. (J LAB CLIN MED 1995;126:81-7)

Abbreviations: CRA = chronic respiratory acidosis; iPTH = immunoreactive parathyroid hormone; $1,25(OH)_2D$ = 1,25 dihydroxyvitamin D; Pa_{CO_2} = partial pressure of carbon dioxide in arterial blood; 25(OH)D = 25-hydroxyvitamin D

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Hypercalciuria is a consistent feature of chronic metabolic acidosis having been found in rats,[1] dogs,[2] and human beings.[3] The increase in urinary calcium excretion is due, in part, to a reduction in the tubular reabsorption of calcium, as augmented urinary calcium excretion is observed in the presence of a decline in the filtered load. The source of the increased calcium that exits into the urine remains unclear, although enhanced calcium removal from bone appears to contribute to the hypercalciuria.[4-6] Intestinal calcium absorption during metabolic acidosis has variably been reported as unchanged[7,8] or increased.[9] In all cases, however, net calcium balance was negative. As a consequence, dissolution of bone and the development of chronic bone disease may occur during chronic metabolic acidosis.[10-12]

In contrast to our knowledge of chronic metabolic acidosis, little is known about changes in calcium metabolism during CRA. Recent data from our laboratory obtained in dogs showed that CRA, similar to chronic metabolic acidosis, is associated with an increment in urinary calcium excretion.[13] That study did not, however, address gastrointestinal calcium handling or net calcium balance during CRA. To examine these issues, we performed complete metabolic balance studies in rats subjected to prolonged hypercapnia. The results demonstrate that CRA is associated with increased urinary calcium excretion and augmented intestinal absorption of calcium such that an increase in net positive calcium balance occurs.

METHODS

Animals. Studies were carried out in male Sprague-Dawley rats weighing between 190 and 250 gm and housed in individual metabolic cages. Standard rat chow containing (by weight) 0.8% calcium and 0.5% phosphorus was employed throughout the study. Rats were given free access to deionized water. Alternating 12 hour light/dark cycles were used, and the temperature of the animal vivarium was maintained at 20° C.

Balance technique. Rats comprising the experimental group were offered 20 gm of diet each day. Those comprising the control group were pair-fed with their experimental group mates. Standard metabolic balance techniques were used throughout the study. Urine specimens were collected daily under mineral oil with thymol as a preservative. Each daily urine collection was acidified with 0.25 ml of 12N HCl before calcium determination. Net calcium balance was calculated as intake minus the combined urine and fecal excretion.

Experimental protocol. Both the experimental and the control group ingested the above described diet under room air conditions for 7 days to allow for adaptation. At the end of this 1-week period, the experimental group (n = 7) was placed in a large environmental chamber containing a 10% CO_2 atmosphere and remained there for a period of 2 weeks. The CO_2 content of the chamber was monitored continuously, and the O_2 content of the atmosphere was maintained at 21%. The control group (n = 7) continued to be observed under room-air conditions during this time period. At the end of this 2-week period all animals were anesthetized with pentobarbital sodium and bled by abdominal aortic puncture within 10 minutes after the administration of the anesthetic agent; for the experimental group, this procedure was carried out while the animals were still maintained in the 10% CO_2 atmosphere. Plasma was obtained for determination of acid-base values, total calcium, iPTH, 25(OH)D, and $1,25(OH)_2D$ levels. Endogenous creatinine clearance was calculated by using the mean urinary creatinine excretion during the last 4 days of experimental week 2 and the plasma creatinine concentration obtained on the last day of the study.

Bone histomorphometry. Femur bone samples were processed for analysis of static bone changes after CRA. Diaphyseal cortical bone was chosen for this analysis based on the assumption that if CRA had a significant impact on mineral metabolism such as occurs during calcium deficiency,[14] marked changes in bone cell activity in the cortical bone would be anticipated. The bones were fixed in 10% neutral buffered formalin for 2 days and demineralized in 5% formic acid with 5% formalin for 3 days, both at 5° C. The demineralized bones were further processed for glycol methacrylate embedding with a Polysciences JB-4 embedding kit.[15] Cross-sections (5 μm) were cut at mid-diaphysis, stained for tartrate-resistant acid phosphatase activity, and counterstained with methyl green–thionin for cell morphology.[15] Osteoclasts were identified by tartrate-resistant acid phosphatase stain and osteoblasts by central negative Golgi area, eccentric nucleus, and strong basophilic cytoplasmic stain.

Analytic methods. Total CO_2 and creatinine were measured by the Technicon Auto Analyzer (Technicon Instruments Corp., Tarrytown, N.Y.). pH was measured anaerobically at 39° C by glass electrode (Radiometer Co., Copenhagen, Denmark). Bicarbonate concentration and $Paco_2$ were calculated from the Henderson-Hasselbalch equation. pH, pK′, and the solubility coefficient of CO_2 were corrected for temperature; pK′ was also corrected for pH.[16-18] Urine calcium and plasma total calcium concentrations were measured by colorimetric assay (Technicon Instruments Corp.). iPTH was measured by radioimmunoassay with a rat PTH standard and antibody recognizing the middle region of the rat PTH molecule (Immuno Nuclear Corp., Stillwater, Minn.). 25(OH)D and $1,25(OH)_2D$ levels were measured by radioimmunoassay.[19]

Statistics. All values are reported as the mean ± SEM unless otherwise specified. Statistical analyses were performed by using a paired or unpaired Student's *t* test where appropriate. Throughout the text the terms *significant* or *significantly different* are used to describe a difference that has a *p* value of less than 0.05, unless otherwise stated.

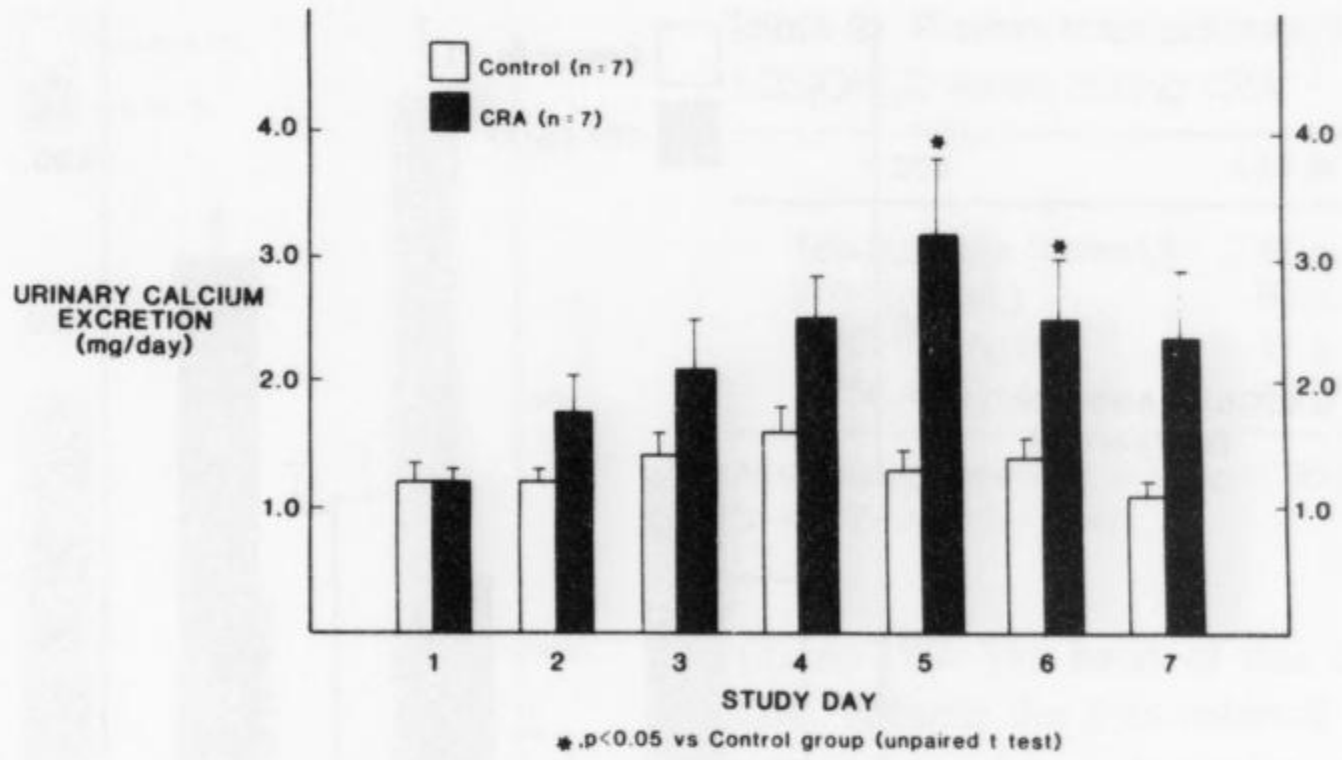

Fig. 1. Urinary calcium excretion during CRA.

RESULTS

Plasma acid-base measurements. The mean values for pH, Pa_{CO_2}, and bicarbonate concentrations obtained at the end of the 2-week study are shown in Table I. All values for the CO_2 group were significantly different from those of the control group and indicate the presence of CRA.

Table I. Plasma acid-base measurements during CRA

	CRA (n = 7)	Control (n = 7)
pH	7.33 ± 0.01*	7.47 ± 0.01
Pa_{CO_2} (mm Hg)	83 ± 4.0*	35 ± 2.0
HCO_3^- (mEq/L)	43 ± 1.0*	25 ± 1.0

All values are means ± SEM.
*$p < 0.05$ vs control group.

Calcium balance

Dietary calcium intake. There were no significant differences in dietary calcium intake between the two groups throughout the study (Table II). Cumulative mean calcium intake over the 2-week period of hypercapnia was 1990 ± 61 mg for the CO_2 group and 1934 ± 54 mg for the control group. Similarly, no significant differences in cumulative weight gain during the 2-week experimental period were observed, mean values being 23 ± 5 gm and 36 ± 5 gm for the CO_2 and control groups, respectively.

Urinary calcium excretion. Mean daily urinary calcium excretion in the CO_2 group increased significantly on days 5 and 6 of hypercapnia (Fig. 1). Mean urinary calcium excretion was significantly higher in the CO_2 group as compared with controls during both the first and second weeks of hypercapnia (16 ± 3 mg vs 9 ± 1 mg, week 1; 16 ± 2 mg vs 9 ± 1 mg, week 2; CO_2 group vs controls, respectively [Table II]). Cumulative mean urinary calcium excretion for the CO_2 group during hypercapnia was almost two times higher than that of the control group (32 ± 5 mg vs 18 ± 2 mg, respectively [$p < 0.05$]).

Endogenous creatinine clearance was similar in the two groups (0.03 ± 0.0002 ml/sec vs 0.03 ± 0.001 ml/sec, CO_2 group vs controls, respectively).

Fecal calcium excretion. Mean fecal calcium excretion in the CO_2 group throughout the period of hypercapnia was not significantly different from that of the control group, although a trend toward decreased excretion was evident (Table II).

Net intestinal calcium absorption. Mean net intestinal calcium absorption (calculated as intake minus fecal excretion) was increased significantly throughout the period of hypercapnia (Fig. 2) (week 1, 213 ± 19 mg vs 135 ± 15 mg [$p < 0.01$]; week 2, 135 ± 16 mg vs 43 ± 14 mg [$p < 0.01$]; cumulatively, 344 ± 27 mg vs 178 ± 20 mg [$p < 0.01$]; CO_2 group vs controls, respectively).

Net calcium balance. As a consequence of the marked increment in intestinal calcium absorption, and despite the hypercalciuria during hypercapnia, mean net calcium balance was significantly more positive in the CO_2 group when compared with the control group throughout the period of hypercapnia (Table II, Fig. 3).

Plasma total calcium, iPTH, 25(OH)D, and $1,25(OH)_2D$. There were no significant differences between the CO_2 and control groups for mean values of plasma total calcium, iPTH, or 25(OH)D (Table III). However, the mean plasma $1,25(OH)_2D$ level for the CO_2 group, 45 ± 6 pg/ml, was significantly lower than the level of 82 ± 9 pg/ml measured in the control group.

Bone histomorphometry. There were no significant effects of CRA on periosteal or endosteal perimeter or

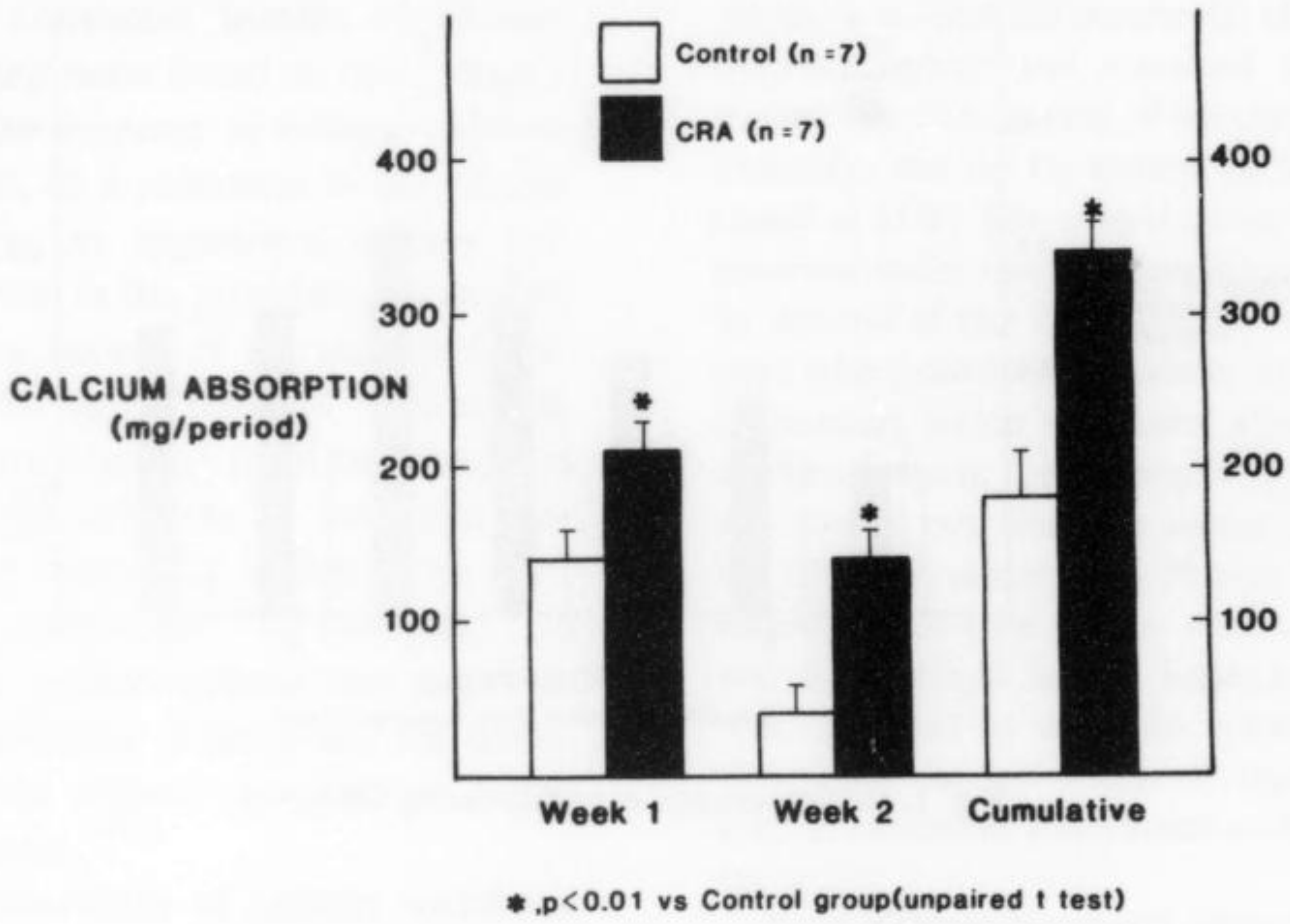

Fig. 2. Net intestinal calcium absorption during CRA.

Table II. Calcium balance during CRA

	Week 1		Week 2		Cumulative	
	CRA	Control	CRA	Control	CRA	Control
Dietary intake	1013 ± 49	991 ± 49	977 ± 38	943 ± 32	1990 ± 61	1934 ± 54
Urinary excretion	16 ± 3*	9 ± 1	16 ± 2*	9 ± 1	32 ± 5*	18 ± 2
Fecal excretion	800 ± 46	857 ± 40	842 ± 31	901 ± 27	1642 ± 48	1757 ± 53
Net balance	197 ± 18†	126 ± 15	120 ± 15†	34 ± 15	317 ± 25†	159 ± 20

All values are expressed as mean ± SEM, mg calcium per period.
*$p < 0.05$, CRA (n = 7) vs control group (n = 7).
†$p < 0.01$, CRA (n = 7) vs control group (n = 7).

in total area (cortical bone area plus medullary area) (Table IV). However, medullary area was significantly increased (4.88 ± 0.32 mm^2 vs 4.07 ± 0.15 mm^2, $p < 0.05$, CO_2 group vs controls). Although cortical bone area did not change significantly in CRA, a small but significant decrease was obtained when correction for the difference in bone size was applied—that is, cortical bone area/total area (0.529 ± 0.017 vs 0.586 ± 0.011, $p < 0.05$, CO_2 group vs controls). CRA did not significantly alter the numbers of osteoclasts and osteoblasts at the endosteum (Table IV), but the number of osteoclasts at the periosteum was significantly increased (0.47 ± 0.07/mm vs 0.16 ± 0.06/mm, $p < 0.05$, CO_2 group vs control).

DISCUSSION

Several interesting findings were observed after a 2-week exposure of rats to a 10% CO_2 atmosphere. First, hypercalciuria was noted during the first week of hypercapnia that persisted throughout the second week of observation and occurred in the absence of a significant change in plasma total calcium, iPTH, or creatinine clearance. Second, net intestinal absorption of calcium was increased during chronic hypercapnia, despite a significant reduction in plasma $1,25(OH)_2D$ levels. Third, the observed increase in net intestinal absorption of calcium during chronic hypercapnia resulted in greater calcium retention as compared with pair-fed room-air controls, despite the hypercalciuria accompanying the hypercapnic exposure.

Chronic metabolic acidosis consistently results in hypercalciuria in human beings[3] and in several animal species.[1,2,20] Micropuncture studies in the dog[2] and measurements of glomerular filtration rate in human beings[3] during ammonium chloride–induced chronic metabolic acidosis have demonstrated a reduction in net tubular calcium absorption as the mechanism of hypercalciuria. Studies of the impact of CRA on urinary calcium excretion are limited, but with one exception[13] they have failed to demonstrate hypercalciuria.[21-23] Lau et al.[23] recently investigated the effect of a 4-day exposure to 10% CO_2 atmosphere on urinary calcium excretion in rats. They were unable to

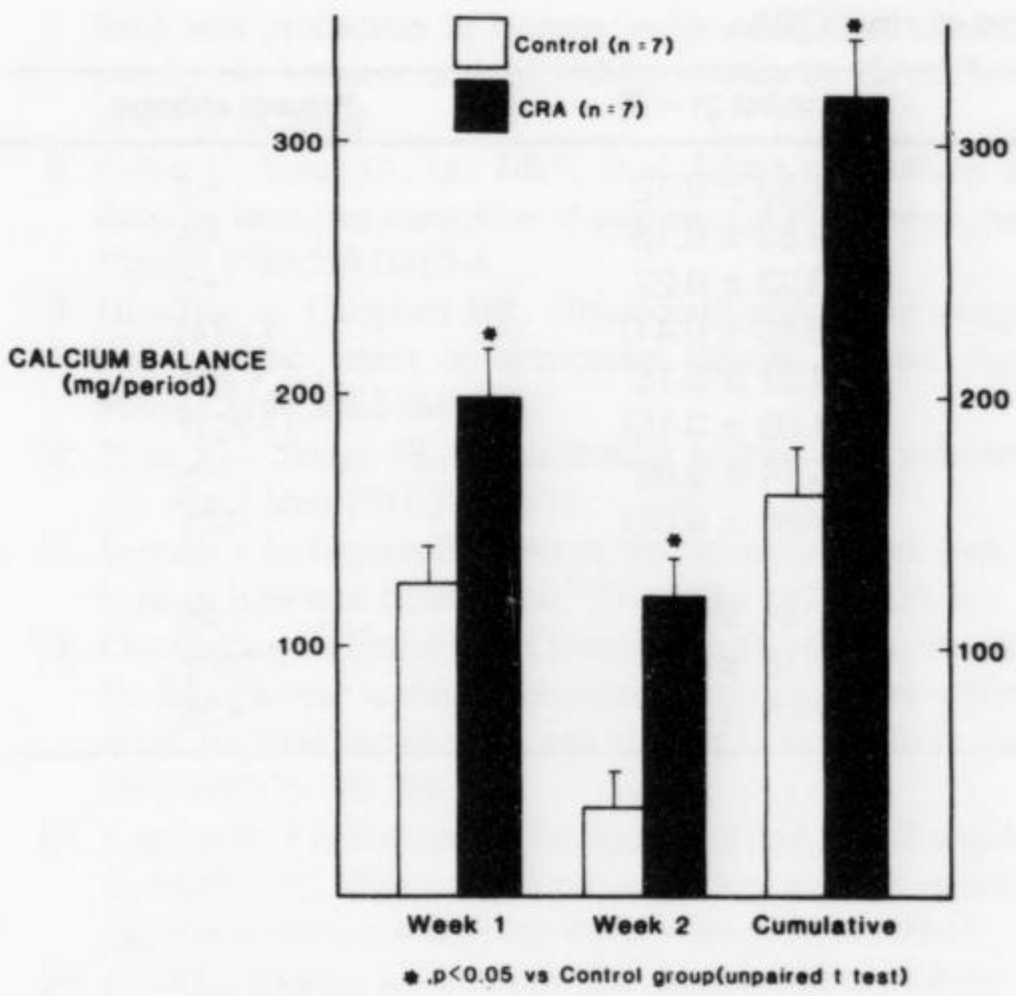

Fig. 3. Net calcium balance during CRA.

Table III. Plasma total calcium, iPTH, 25(OH)D, and $1,25(OH)_2D$ levels during CRA

	CRA (n = 7)	Control (n = 7)
Total calcium (mmol/L)	2.42 ± 0.02	2.37 ± 0.07
iPTH (pmol/L)	69 ± 5	62 ± 4
25(OH)D (ng/ml)	17 ± 1	17 ± 1
$1,25(OH)_2D$ (pg/ml)	45 ± 6*	82 ± 9

All values are expressed as mean ± SEM.
*$p < 0.05$ vs control group.

demonstrate any daily or cumulative change in urinary calcium excretion as compared with pair-fed controls breathing room air. In this regard it is interesting to note that significant hypercalciuria was not observed in our study until day 5 of hypercapnia. It is possible that such a time period is necessary before urinary calcium excretion is enhanced, although our findings suggest a trend toward hypercalciuria as early as day 2 of hypercapnia (Fig. 1). In accord with our present observations in rats, in previous studies from our laboratory in dogs maintained in a 10% CO_2 atmosphere for 2 weeks, significant hypercalciuria was observed during the first week of hypercapnia that persisted throughout the second week of study.[13] We therefore conclude that CRA, just like chronic metabolic acidosis, leads to an augmentation of urinary calcium excretion.

The mechanism of hypercalciuria observed in the present study is unclear. Based on the static bone parameters obtained (Table IV), especially at the endosteum, the bone site where substantial changes in bone cell activity might be expected,[14] it is evident that no significant changes in the number of endosteal osteoclasts or osteoblasts were observed. These results suggest that increased bone resorption or decreased bone formation at the endosteum does not contribute to the hypercalciuria induced by CRA. This suggestion is consistent with the absence of a significant change in the plasma level of iPTH during chronic respiratory acidosis (Table III). Hypercapnia was attended by a mild, although significant, increase in bone medullary area irrespective of whether the intracortical spaces were included in the measurement (Table IV). The basis of this change remains uncertain, because the data referred to above did not support increased bone resorption or decreased bone formation at the endosteum. The trend toward greater bone size (as indicated by total area) in CRA might account, at least in part, for this change, but analysis of dynamic bone changes at the endosteum will be required to explore further the nature of these findings. Cortical bone area remained unchanged in chronic hypercapnia, although a marginal, albeit significant, decrease in the cortical bone area/total area index was observed. Again, the trend toward an increase in total area might be responsible for this change. Finally, the number of periosteal osteoclasts was increased in CRA, but this change was not accompanied by a significant decrease in total area. Because bone changes at the periosteum are normally associated with growth and modeling rather than mineral homeostasis,[24] the significance of this cell change remains to be established.

The absence of a significant difference in plasma total calcium and creatinine clearance between the CO_2 and control groups suggests that the hypercalciuria is not the result of an increased filtered load of calcium. Measurement of plasma ionized or ultrafiltrable calcium and glomerular filtration rate (e.g., inulin clearance), variables that were not determined in the present study, will be needed to clarify this issue. As noted, previous studies have established reduced net tubular calcium absorption as the mechanism of hypercalciuria associated with chronic metabolic acidosis.[2,3] We have previously observed an increase in fractional excretion of calcium in the absence of changes in iPTH or $1,25(OH)_2D$ during CRA in the dog.[13]

Net intestinal calcium absorption increased during CRA and resulted in a significantly more positive calcium balance in the CO_2 group. This finding is in sharp contrast to observations during experimental[8] and clinical[7] states of chronic metabolic acidosis in which net intestinal calcium absorption fails to compensate for hypercalciuria, thus resulting in progressively negative calcium balance. In addition, iPTH

Table IV. Static bone parameters in the mid-femur cross-section during CRA

Parameter	CRA (n = 7)	Control (n = 7)	Percent change
Periosteal perimeter (mm)	12.12 ± 0.20	11.83 ± 0.12	+2
Total area (mm^2)	10.40 ± 0.31	9.83 ± 0.13	+6
Endosteal perimeter (mm)	8.67 ± 0.46	8.23 ± 0.22	+5
	(10.01 ± 0.74)*	(8.80 ± 0.41)	(+14)
Medullary area (mm^2)	4.88 ± 0.32	4.07 ± 0.15	+20†
	(4.92 ± 0.32)*	(4.09 ± 0.15)	(+20)†
Cortical bone area (mm^2)	5.48 ± 0.13	5.74 ± 0.09	−5
Cortical bone area/total area	0.529 ± 0.017	0.586 ± 0.011	−10†
Osteoclasts/mm			
Endosteal	0.81 ± 0.22	1.48 ± 0.44	−45
Periosteal	0.47 ± 0.07	0.16 ± 0.06	+194†
Endosteal osteoblasts/mm	30.4 ± 3.8	23.8 ± 3.5	+28

All values are expressed as mean ± SEM unless otherwise specified.
*Including length or area measurements in intracortical spaces.
†$p < 0.05$ vs control group.

and vitamin D levels are typically normal during chronic metabolic acidosis,[2,3,8,25,26] although in vitro studies have demonstrated impaired 1 α-hydroxylase activity during acidemic states.[27,28] A reduction in the activity of this enzyme during the relatively severe CRA produced in the present study might account for the significant reduction in $1,25(OH)_2D$ level that we observed.

The mechanism of the enhanced net intestinal absorption of calcium that occurs during CRA despite a decreased plasma $1,25(OH)_2D$ concentration is unclear. One possibility is a vitamin D–independent mechanism of absorption. Increased intestinal calcium absorption occurring independent of changes in plasma $1,25(OH)_2D$ has been reported in rats during bicarbonate feeding[29] and furosemide administration.[30] Conversely, increased plasma $1,25(OH)_2D$ in the absence of changes in both net intestinal calcium absorption or duodenal gut sac calcium uptake has also been reported during chronic metabolic acidosis.[31]

The hypercalciuria observed during CRA accounted for only a small fraction of the total amount of calcium retained by each rat in the CO_2 group. An important issue is the ultimate destination of this retained calcium. Bone mineral content was not examined during this study, although preliminary findings in chronically hypercapnic dogs indicate the presence of increased bone calcium content.[32] In neonatal mouse calvariae, decreased bone carbonate content has recently been demonstrated in response to metabolic, but not respiratory, acidosis of 48 hours' duration.[33] Additional studies in the neonatal mouse calvariae model have demonstrated net calcium efflux from live bone during chronic metabolic, but not chronic respiratory, acidosis.[34] These findings stand in sharp contrast to the known bone dissolution and the potential for development of chronic bone disease in association with chronic metabolic acidosis.[10-12] Furthermore, preliminary observations suggest that exposure of rat osteoblast-like cells to respiratory acidosis produces a significantly greater increase in intracellular calcium concentration and decrease in intracellular pH as compared with metabolic acidosis.[35] Whether these findings are responsible for the different bone calcium response to the two acidemic acid-base disorders is not known.

In summary, CRA in the rat produces intestinal hyperabsorption of calcium and augmented calcium retention despite accompanying hypercalciuria. The mechanisms of the observed intestinal and renal changes during chronic hypercapnia and their impact on bone metabolism require further study.

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